MEDIUM · INFORMATION AND IDEAS · ANSWER KEY

Information and Ideas

02

10 answers · Rationale-rich · Teach from doc

Q1 D

D. Buck holds Thornton in higher regard than any other person.

Choice D is the best answer because it most accurately states the main idea of the text. After establishing that Buck views most people "as nothing," the text explains that Buck won't acknowledge people other than Thornton unless they appear friendly toward Thornton, and even then he's only reluctantly accepting. Thus, the text focuses on the idea that Thornton has a special status in Buck's mind, with Buck holding him in higher regard than other people.

Choice A is incorrect because the text conveys that Buck isn't social with people other than Thornton but doesn't address Buck's life or temperament before he lived with Thornton. Choice B is incorrect because the text conveys that Buck doesn't really care about people other than Thornton and is aloof toward them. However, there's no indication that Buck mistrusts and avoids people generally; indeed, he accepts Thornton, who is a human. Choice C is incorrect because the text refers to random travelers praising and petting Buck and Thornton's partners giving Buck favors, but there's no indication that any of these people are Thornton's friends or that they have a particular fondness for Buck.

Q2

A

A. identify molecular components similar to EBF that target the activation of Or31 receptors.

Choice A is the best answer because it most logically completes the text's discussion of mosquito repellents. The text begins by explaining that many repellents work by using natural components to activate multiple odor receptors on mosquitoes' antennae, and that new repellents must be created whenever mosquitoes become resistant to older ones. The text then highlights a research team's discovery that EBF, a molecular component of a chrysanthemum-flower extract, can repel mosquitoes by activating a single odor receptor, Or31, that is shared by all species of mosquitoes known to carry diseases. The text suggests that compared to the repellents mentioned earlier, a repellent that acts on the Or31 receptor would be more effective: by noting that all mosquito species known to carry diseases share the Or31 receptor, the text suggests that the Or31 receptor may be unique in this respect, meaning that a repellent such as EBF that acts on it would be more effective since it works on a single receptor shared by all mosquito species that carry diseases, rather than a combination of receptors that is not shared by all species. Once mosquitoes become resistant to EBF, it would therefore make sense for researchers to look for other molecular components similar to EBF that target the activation of Or31 receptors, since a single such component could also repel all disease-carrying mosquitoes.

Choice B is incorrect because nothing in the text suggests that EBF molecules are difficult to extract from chrysanthemums and that investigating alternative extraction methods would therefore be useful for developing efficient and effective mosquito repellents. Rather, the text suggests that researchers developing new mosquito repellents should aim to identify molecular components similar to EBF, since that component targets the Or31 odor receptor shared by all species of mosquitoes known to carry diseases. Choice C is incorrect because nothing in the text suggests that researchers are unaware of the precise location of Or31 and other odor receptors in mosquitoes' antennae or that knowing this information would be useful for developing efficient and effective mosquito repellents. Rather, the text suggests that researchers developing new mosquito repellents should aim to identify molecular components similar to EBF, which targets the Or31 odor receptor. Choice D is incorrect because it doesn't logically follow that the discovery of one odor receptor shared by all disease-bearing mosquitoes should lead to further research into which repellents might activate the greatest number of odor receptors. Rather, the text suggests that researchers developing new mosquito repellents should instead search for additional molecular components that, like EBF, activate the one odor receptor that is known to be shared by all disease-bearing mosquitoes.

Q3

D

D. Each of the planets has a mass greater than 0.25 Jupiters, and all except for TOI-1478 b have an orbital period of less than 10 days.

Choice D is the best answer because it accurately describes data from the table that support Rodriguez and colleagues' assertion about the classifications of the five new gas exoplanets. The text describes two categories of gas planets: hot Jupiters, which have a mass of at least 0.25 Jupiters and an orbital period of less than 10 days, and warm Jupiters, which have the same mass characteristic but have orbital periods of more than 10 days. According to the table, four of the gas exoplanets discovered by Rodriguez and colleagues have a mass of at least 0.25 Jupiters and an orbital period of less than 10 days, while one of the planets has a mass of at least 0.25 Jupiters and an orbital period of more than 10 days. These data therefore support Rodriguez and colleagues' assertion that four of the new exoplanets are hot Jupiters and one is a warm Jupiter.

Choice A is incorrect because it doesn't accurately describe the data from the table. Although the table shows that TOI-628 b has a mass equivalent to 6.33 Jupiters, the table also shows that one of the planets—TOI-1478 b—does indeed have an orbital period of more than 10 days. Choice B is incorrect because it doesn't accurately describe the data from the table. Although the table does show that the masses of the five planets range from 0.85 to 6.33 Jupiters, the table also shows that TOI-1478 b has an orbital period of 10.180 days, not 153 days. Choice C is incorrect. According to the table, TOI-1333 b has an orbital period of only 4.720 days, not more than 10 days. Additionally, although the table does show that all the planets have a radius between 1.060 and 1.771 Jupiters, the text indicates that a planet may be classified as a hot Jupiter or a warm Jupiter based on its mass and orbital period, not on its radius, making the information about the range of the five planets' radius values irrelevant.

Q4

A

A. Historians agree that Mieszko II Lambert existed, but disagree about whether Siemomysł existed.

Choice A is the best answer because it presents a statement about how historians view Siemomysł and Mieszko II Lambert that is supported by the text. The text states that the Piast dynasty had a number of different members. The text refers to two of the rulers in the Piast dynasty by name: Mieszko II Lambert, whose rule was known to have occurred from 1025 to 1031 CE, and Siemomysł ("whose historical actuality is disputed"), for whom less is known and who therefore is the subject of debate among historians. The text further casts doubt about Siemomysł by stating that he is "said to have ruled" during the 10th century, or the 900s, which suggests the possibility that he didn't rule. The text also mentions that the chronicle's documentation of Siemomysł relies on oral tradition, unlike its records of later rulers, including Mieszko II Lambert. This indicates that historians agree that Mieszko II Lambert was an actual historical figure, but they disagree about whether Siemomysł existed.

Choice B is incorrect because the text states that the *Gesta principum Polonorum*'s documentation of Siemomysł comes from oral tradition—spoken rather than written documentation—unlike its records of rulers who came after Siemomysł. This suggests that the chronicle provides less reliable evidence rather than more evidence for Siemomysł's existence in the 900s (or the 10th century) than it does for later rulers like Mieszko II Lambert, who ruled from 1025 to 1031 CE. Choice C is incorrect because the text indicates the opposite: Siemomysł supposedly ruled earlier in the 10th century, while Mieszko II Lambert ruled from 1025 to 1031 CE. The 10th century spans the years from 901 to 1000 CE, so Siemomysł is believed to have ruled earlier than Mieszko II Lambert did, not later as this choice states. Choice D is incorrect. Although the text mentions that information in the written chronicle *Gesta principum Polonorum* draws its information about Siemomysł from the oral tradition, it doesn't mention orally transmitted stories about Mieszko II Lambert. Instead, it states that the chronicle's documentation of Siemomysł relies on oral tradition unlike the records of later rulers do, which suggests that the documentation of later rulers such as Mieszko II Lambert did not rely on oral tradition. Thus, no comparison can be made about how convincing the orally transmitted stories of each ruler are to historians.

Q5

B

B. the sound quality of Cremonese violins results in part from a method the craftspeople used to alter the wood.

Choice B is the best answer because it most logically completes the text's discussion of the sound quality of Cremonese and modern violins. The text states that violins made in Cremona in the sixteenth to eighteenth centuries sound superior to modern violins. It then indicates that some experts attribute the difference to different woods being used to make these violins, but both Cremonese and modern violins are made of the same woods (maple and spruce); thus this cannot account for the difference. The text then says that recent analysis suggests the wood in Cremonese violins was chemically treated by the craftspeople who made them, thereby providing an alternate explanation for the sound differences: the chemical alteration that is present in the Cremonese violins but absent from the modern ones.

Choice A is incorrect because the text does not discuss how the sound quality of Cremonese violins compares with the sound quality of other instruments made during the sixteenth to eighteenth centuries. Instead it focuses on how the sound of the Cremonese violins compares with that of modern violins. Choice C is incorrect. The text states that there are differences in sound quality between the Cremonese and modern violins, and that both types of violin are made with maple or spruce. Thus the type of wood alone does not determine a violin's sound quality. Furthermore, even if the type of wood alone could account for differences in sound quality, the text makes no mention of other woods, so there is no basis to judge how modern violins would sound if they were made using woods besides maple and spruce. Choice D is incorrect because the text states that there is evidence that Cremonese craftspeople chemically treated the wood used in Cremonese violins. This evidence is attributed to "new analysis," which strongly suggests that this process was unknown to modern violin makers before that analysis. If the chemical treatment was unknown until recently, the manufacturing process for modern violins must differ with respect to the previously unknown practice of chemically treating the wood.

Q6

D

D. 3.6% of respondents said it was most effective to use efficient light bulbs.

Choice D is the best answer because it most effectively uses data from the table to complete the text's discussion of Attari and her team's survey results. The text states that the team asked respondents to identify the most effective action people can take to save energy, with the team classifying each action as either an efficiency or a curtailment. According to the text, respondents named curtailments more often than they did efficiencies. The text then offers an example that begins by citing a curtailment, turning off the lights, that was selected by a relatively high percentage of respondents (19.6%). Given that the example is presented in support of the idea that more respondents selected curtailments than efficiencies, the most effective way to complete the example is by citing an efficiency, using efficient light bulbs, that was selected by a relatively low percentage of respondents (only 3.6%).

Choice A is incorrect because it inaccurately describes data in the table. The data indicate that 6.3% of respondents said the most effective action was to change the thermostat setting, not to use efficient cars or hybrids. Choice B is incorrect because it inaccurately describes data in the table. The data indicate that 2.8% of respondents said the most effective action was to use efficient cars/hybrids, not to change the thermostat setting. Choice C is incorrect because it mentions a curtailment (using a bike or public transportation) and not an efficiency. The text states that a research team asked respondents to identify the most effective action people can take to save energy, with the team classifying each action as either an efficiency or a curtailment. According to the text, respondents named curtailments more often than they did efficiencies. The text then offers an example that begins by citing a curtailment, turning off the lights, that was selected by a relatively high percentage of respondents (19.6%). Given that the example is presented in support of the idea that more people selected curtailments than efficiencies, the most effective way to complete the example is not by referring to another curtailment but rather by referring to an efficiency that was selected by a relatively low percentage of respondents.

Q7 D

D. The Aspera mission is expected to produce the first direct observations of CGM gas in the “warm-hot” phase, which likely has an important role in the evolution of galaxies.

Choice D is the best answer because it most accurately states the main idea of the text. The text begins by mentioning NASA’s Aspera mission, which will investigate the low-density gas that makes up the circumgalactic medium (CGM). According to the text, this mission will focus on a portion of the CGM’s gas that exists in a “warm-hot” phase; this “warm-hot” gas has not been previously observed, but it is thought to make up most of the mass of galaxies and play a part in star formation. Finally, the text mentions a telescope capable of examining this previously unobservable “warm-hot” gas: the Aspera mission will use this telescope in the hope of answering questions about galaxy formation and change. Therefore, the main idea of the text is that the Aspera mission is likely to produce the first direct observations of CGM gas in the “warm-hot” phase, which likely has an important role in the evolution of galaxies.

Choice A is incorrect. Although this choice mentions the Aspera mission, names its leader, and generally states the mission’s purpose, it does not reference the “warm-hot” gas or fully convey the reason why the Aspera mission is significant. Choice B is incorrect. Although this choice mentions the “warm-hot” gas that makes up a portion of the CGM, it does not reference the Aspera mission or describe its importance. The text also does not mention that galaxies surrounded by the CGM have been studied. Choice C is incorrect. Although this choice describes a problem related to the CGM that researchers have been attempting to solve and presents the speculation of those researchers, it does not mention the Aspera mission or describe its purpose.

Q8

C

C. more narwhals would have a tusk.

Choice C is the best answer because it most logically completes the text's discussion of Kristin Laidre's reasoning about the purpose of the tusk that many, but not all, narwhals have. The text explains that one group of scientists thinks the tusk may help narwhals detect the threat of freezing water and that Laidre disagrees with that idea, given the importance of avoiding a dangerous situation. It's logical to suggest that if the tusk serves such an important purpose for narwhals, the trait would be more common among them—specifically, that more narwhals would have a tusk.

Choice A is incorrect because there's no reason to think Laidre would say that if the tusk has the important function of helping narwhals detect when the water around them is about to freeze (meaning that it isn't always freezing), some narwhals would choose a different habitat altogether. Indeed, if it's true that the tusk helps narwhals avoid areas with dangerous conditions when they occur in their Arctic Ocean habitat, the tusk would likely enable the narwhals to continue living in that habitat rather than drive them elsewhere entirely. Choice B is incorrect because the text focuses only on narwhals and makes no mention of other marine animals or how having a tusk might affect them. And if anything, it would be more logical to expect a very important trait to be more widespread, not less common, among other similar types of animals. Choice D is incorrect. Although the text describes narwhals as shy, it doesn't indicate that the scientists' conclusion has anything to do with shyness. And because shyness and detection of the threat of freezing water aren't logically connected, there's no reason to think that Laidre would expect narwhals to become less shy over time if the tusk serves that important purpose.

Q9

B

B. “Their taste was strikingly alike. The same books, the same passages were idolized by each—or if any difference appeared, any objection arose, it lasted no longer than till the force of her arguments and the brightness of her eyes could be displayed.”

Choice B is the best answer. By showing that "any difference" in taste was quickly overcome by "the force of [Marianne’s] arguments," this choice effectively demonstrates Marianne’s "ability to persuade others."

Choice A is incorrect. This choice doesn’t effectively illustrate the claim. This choice shows that Marianne and John share an interest in music and dancing, but it doesn’t provide evidence of Marianne’s "ability to persuade others." Choice C is incorrect. This choice doesn’t effectively illustrate the claim. This choice shows that Marianne enjoys talking about her interests, but it doesn’t provide evidence of Marianne’s "ability to persuade others." Choice D is incorrect. This choice doesn’t effectively illustrate the claim. This choice shows that Marianne and John share many interests and generally agree on music and dancing, but it doesn’t provide evidence of Marianne’s "ability to persuade others."

Q10 D

D. All the marsquakes started from the same location on the planet.

Choice D is the best answer because it presents a statement about what surprised the scientists that is supported by the text. The text states that the marsquakes described in the data from NASA's InSight lander originated from the same location on Mars. The text goes on to say that because they had expected the opposite (that marsquakes would originate from all over the planet) this discovery surprised the scientists.

Choice A is incorrect because the text doesn't say that the data from NASA's InSight lander revealed any surprising information about the planet's surface temperature. Instead, the text mentions the cooling of Mars's surface as a reason the scientists expected that marsquakes had multiple origins. In addition, cooling would indicate that the temperature has been falling rather than rising. Choice B is incorrect. Although the text indicates that by studying seismic activity scientists found a possible explanation for what causes marsquakes, the text doesn't say that they discovered that marsquakes are caused by different types of seismic waves. Rather, the text states that based on the data from NASA's InSight lander, scientists now believe that this seismic activity happens because of areas of active magma that flow below the planet's surface. Choice C is incorrect because the text doesn't discuss the amount of data NASA's InSight lander collected or whether scientists who studied the data found the amount to be as expected. Instead, the text focuses on what the data revealed about where on Mars the marsquakes originated.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank